

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC IFoS

Subject: Forestry

Questions: 8

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AI-generated PYQs: 0

1. [UNKNOWN] paper-i

Q2. (a) Describe the natural distribution of *Shorea robusta*, and *Azadirachta indica*. 10 (b) Write silvicultural characters of *Casuarina equisetifolia*, and (ii) *Abies pindrow*. 10 (c) Some rural communities are opposed to chir pine and advocate removal of chir pine and its replacement with broadleaved multipurpose trees. What is your reaction in this matter? 10 (d) Regulation of solar radiation gives a powerful tool to the forester. Justify. 10

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2. [UNKNOWN] paper-i

Q3. Where are cold deserts found in India ? Explain site characteristics encountered in a cold desert and steps suggested to overcome problems in their afforestation. 10 Describe the technique of Sand Dune Fixation in the Thar Desert. Also mention the choice of species for planting. 15 Enlist the general rules governing the felling of bamboo in forests. What are the consequences of gregarious flowering of bamboo ? What special measures would you take in the event of gregarious flowering? 15 FSI-P-FSTY 2 ===== PAGE 3 =====

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3. [UNKNOWN] paper-i

Q4. (a) Describe the unique characteristics of mangrove forest vegetation. 10 A soil can be wet, yet physiologically dry. How ? What steps are suggested to correct the problem? 10 What is Sub-climax? Explain its importance in the context of Indian forestry. 10 Enlist the advantages and disadvantages of vegetative propagation. What future do you foresee for it in forestry? 10 FSI-P-FSTY 3 ===== PAGE 4 ===== SECTION B

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4. [UNKNOWN] paper-i

Q5. Answer the following: 8x5=40 Write in brief on advanced generation tree improvement. 8 Discuss in detail, the kinds of soil mycorrhizae and the benefits derived by plants from them. 8 Explain General Combining Ability (GCA), Specific Combining Ability (SCA) and their utility. 8 Comment on the possible impact of greenhouse gases on the global environment. 8 Discuss the impact of agroforestry practices on the environment in general and on soil properties in particular, with a suitable example. 8

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5. [UNKNOWN] paper-i

Q6. (a) Discuss the term Biodiversity. Explain the levels in which it can be studied. What are the different measures employed to measure biological diversity? Elaborate. 20 (b) Explain the scope and aims of Urban forestry. Discuss the need of urban forestry in the improvement of city environment. 20

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6. [UNKNOWN] paper-i

Q7. (a) What are Seed Production Areas (SPA) ? Explain the purpose of establishing them. Briefly highlight the advantages and disadvantages of SPA. List the steps involved in establishing SPA. 20 What are the various methods adopted to conserve the soil on sloping areas ? Explain in brief. 10 Differentiate between Windbreaks and Shelterbelts. Discuss in brief, their impact on the environment. 10

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7. [UNKNOWN] paper-i

Q8. (a) What is Water harvesting ? List the different methods of water harvesting and suggest various practices for

efficient use of conserved water. 20 Explain various types of Succession. Describe the major theories explaining succession. 10 List the different components of Phenotypic variation. How are they important for tree breeders? 10 FSI-P-FSTY 4

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8. [UNKNOWN] paper-ii

1. Answer each of the following questions: Why site specific planning is essential for forest management ? Explain different components of site-specific management. 8 How the forest cover was measured prior to and post 1980's in India? Define the various categories of forest cover. 8 What are the objectives of working plan? How it is helpful for conservation of biodiversity and natural resource conservation of forests? 8 What are the applications of remote sensing and GIS in the field of forestry and wildlife? 8 How volume tables are classified on the basis of kind of outturn ? Describe briefly. 8 How a selection forest is managed under the felling series? Describe with suitable examples/diagrams. 15 How the geometrical measurements for calculating volume yield are made in (i) Buttressed tree, (ii) Leaning tree on a slope, and (iii) a tree forked at the base? Give thematic presentation. 15 Describe the effect of thinning on volume increment. 10 3.(a) Describe the different formulae used in forest trees for determining increment percent in diameter and volume. 15 1 fsi-d-fsty ===== PAGE 2 ===== Describe the various kinds of maps prepared by the Working Plan Officer. What is their utility and purpose ? 1 5 Describe the dead and live loads and how they are calculated on the roof trusses over buildings? 10 Write about the concept of normal forest and kind of abnormalities which affect the normal growing stock. 1 5 Write in detail as to why Joint Forest Management Policy was initiated and what are its constraints in implementation ? 1 5 Write a note on kinds of enumeration and explain in brief different random sampling techniques used in forest inventories. 1 0 SECTION 'B' Define herbarium and write its significance. What are the steps in establishing herbarium and which method of plant classification is followed in herbarium arrangement? 8 Briefly write about the natural defects observed in wood. 8 What are the major grazing systems and grazing regions in India? Describe briefly, methods to prevent pressure from grazing in forests. 8 Narrate the background and objectives of implementation of national forest policies of India. 8 What is ethnobotany and describe the role of ethnobotany in the Indian systems of medicine. 8 Define succession. Explain different types of succession in detail, citing suitable examples. Discuss various theories of succession. 1 5 What is wood seasoning? Write in detail the aims of wood seasoning and explain salient steps in manufacturing of plywood. 15 Discuss briefly the impacts of humans on forest health. Explain different measures to check forest encroachment. 1 0 What is logging? What are the objects of felling? Briefly write the general rules of felling and basic cuts of tree felling. 1 5 Explain the discounting measures in forest business management with different formulae used in forest economics. 1 5 Explain the classification of forest types in India by Champion and Seth. Enlist major forest types and their group. 10 Describe types of forest fires, their ill-effects and preventive measures. Briefly discuss the role of forest fire on forest ecosystems. 1 5 Discuss in detail the roles and interventions of Government and Private agencies in marketing of forest produce. 1 5 What are the differences between gums and resins? How are gums and resins classified ? Enlist different methods of tapping resins. 1 0 • fsi-d-fsty 2

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